

Agricultural Evapotranspiration and Groundwater Depletion in the Cuyama Valley

Vikesh Dheeriya

March 2026

Groundwater depletion in the Cuyama Valley is accelerating, with the ten fastest-declining wells deepening at 7.6–9.3 ft/yr, exceeding the GSP's 2 ft/yr threshold. OpenET data shows that high late-summer irrigation demand from Truck, Nursery & Berry crops, including carrots, correlates with peak aquifer drawdown. These findings suggest that current extraction rates may not align with SGMA's 2040 sustainability deadline.

Introduction

Groundwater is a primary source of freshwater, yet accelerating agricultural demand and reduced precipitation are depleting aquifers rapidly. A global analysis of 170,000 monitoring wells found that groundwater level declines exceed 0.5 m/yr in arid agricultural regions, with declines accelerating in 30% of the world's aquifer systems over the past four decades (1). California's agriculture depends on groundwater, which supplies 40–60% of the state's water. The Cuyama Valley Groundwater Basin has been pumped at roughly twice its recharge rate since the 1950s, removing 2.1 million acre-feet (AF) from storage and dropping the water table more than 300 ft (2, 3). The 2014 Sustainable Groundwater Management Act (SGMA) mandated that all critically overdrafted basins achieve sustainability by 2040, requiring Cuyama to reduce pumping by two-thirds from 60,000 AF/yr to 20,000 AF/yr (2). A 2019 economic analysis by the Cuyama Basin Groundwater Sustainability Agency (CBGSA) found full Groundwater Sustainability Plan (GSP) implementation would fallow 62% of irrigated acreage and reduce gross agricultural revenue by 63% (4). Even as the GSP was being drafted, the basin's crop portfolio shifted toward more water-intensive perennials. Comparison of the 2014 and 2024 California Department of Water Resources (DWR) crop maps reveal substantial vineyard expansion, including an 850-acre vineyard established by Harvard University's endowment subsidiary Brodiaea Inc. (5). Carrot production represented 49% of the Cuyama basin agricultural revenue in 2017 (4). Truck, Nursery, & Berry crops are the DWR classification that encompasses carrot farming (6). Across California, perennial and specialty crop expansion has been shown to increase irrigation demand, increasing revenue losses under pumping restrictions by up to 30% (7).

Methods

Groundwater depth was obtained from the DWR CASGEM monitoring network for 208 OPTI wells spanning 2000–2024 (8). Measurements were spatially filtered to the central agricultural zone because drawdown is most significant in this region (3). Wells with fewer than 10 years of annual

coverage were excluded, leaving 91 wells. Evapotranspiration (ET) was derived from OpenET, an ensemble of six satellite-based models providing monthly 30-m resolution estimates from 2016 to 2024 in Google Earth Engine (9, 10). Crop classifications were obtained from the DWR i15 2024 Provisional Crop Mapping shapefile (6). Truck, Nursery, & Berry crops and Grain & Hay crops were analyzed separately from other agricultural crops because they were the two most prevalent classes. Monthly net irrigation water requirement (NIWR) was computed as ET minus GRIDMET precipitation between April and October, the standard growing season in California (11, 12).

Results

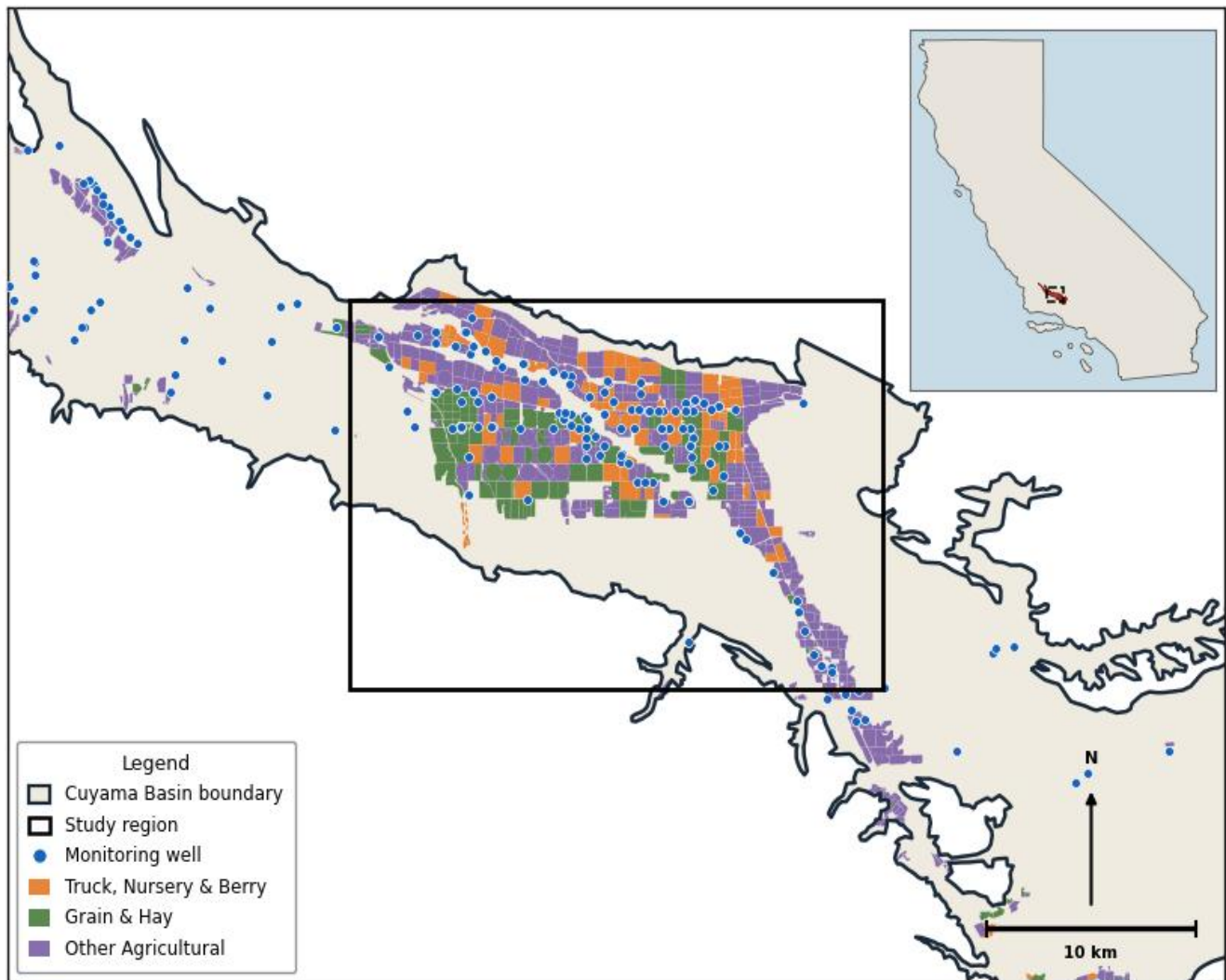


Fig. 1. Study area map of the Cuyama Valley Groundwater Basin. The black box delineates the central agricultural study region. Colored polygons indicate crop classifications from the 2024 DWR i15 Provisional Crop Mapping dataset (6). Orange (Truck, Nursery & Berry), green (Grain & Hay), and purple (Other Agricultural) are shown. Blue dots indicate CASGEM monitoring wells. Inset map shows

basin location within California. Groundwater well locations and measurements from DWR CASGEM (8).

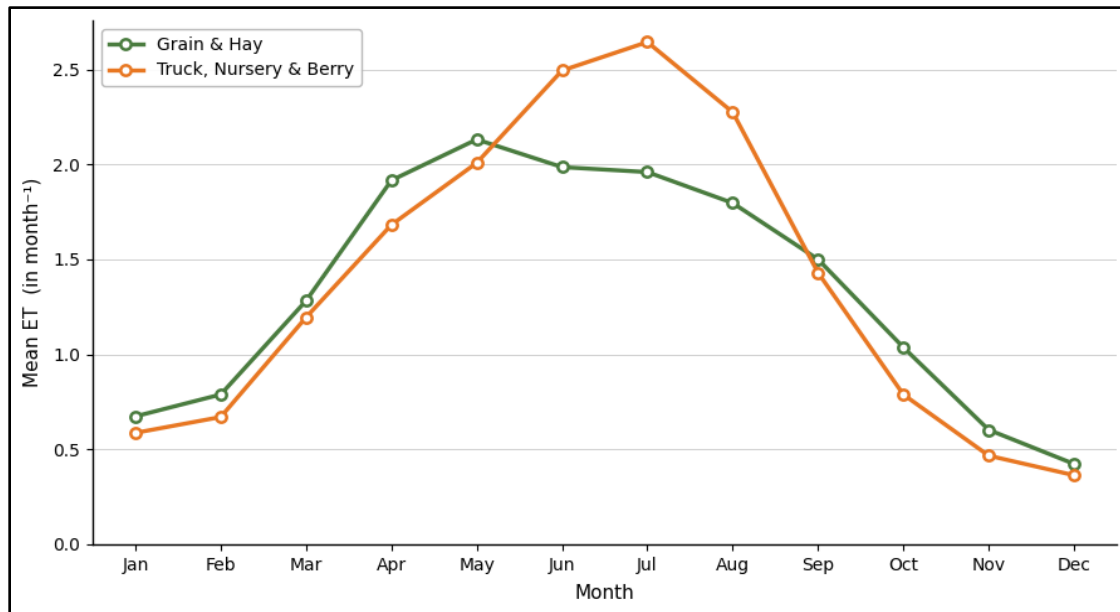


Fig. 2. Mean monthly evapotranspiration (mm month^{-1}) for Grain & Hay (green) and Truck, Nursery & Berry crops (orange) from 2016–2024. Grain & Hay peaks in May while Truck, Nursery & Berry peaks in July. ET data acquired from OpenET ensemble (9).

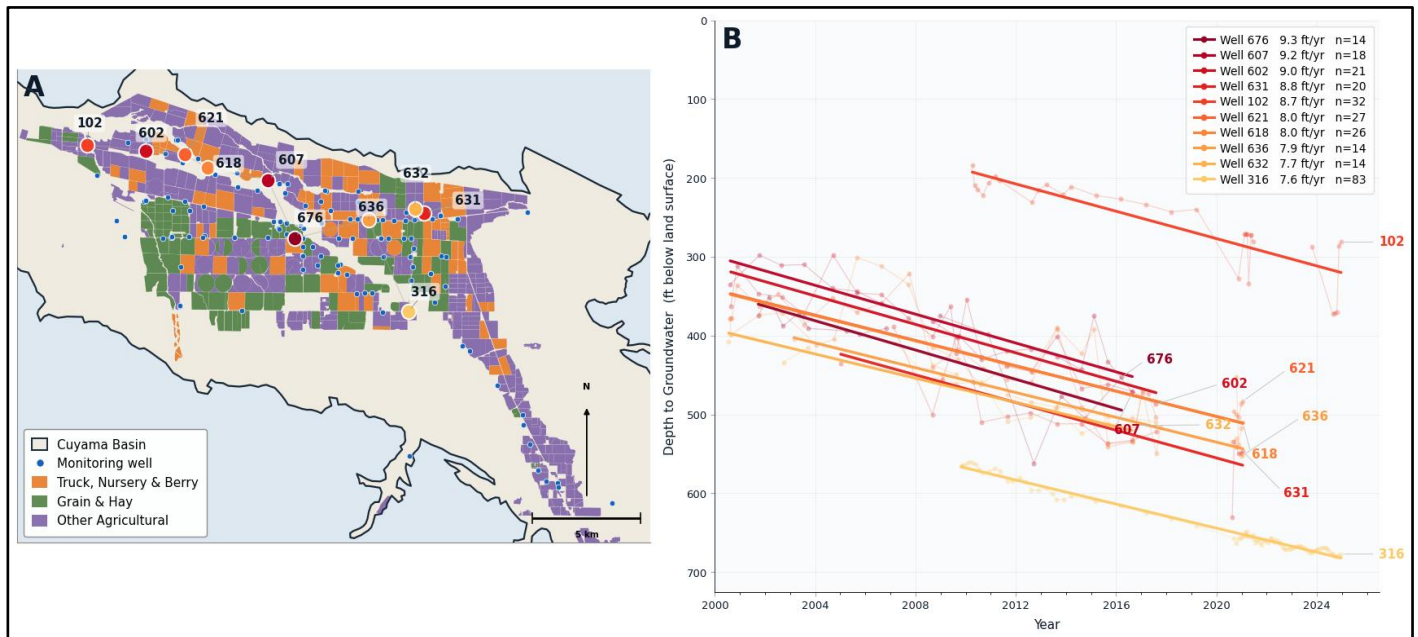


Fig 3. Spatial distribution of the ten fastest depleting wells overlaid on the crop type map (A). Individual well DTGW time series (2000–2024) with trend lines (B). Wells are colored by depletion rate

(ft/yr) from fastest (dark red) to slowest (light orange). Groundwater well locations and measurements from DWR CASGEM (8).

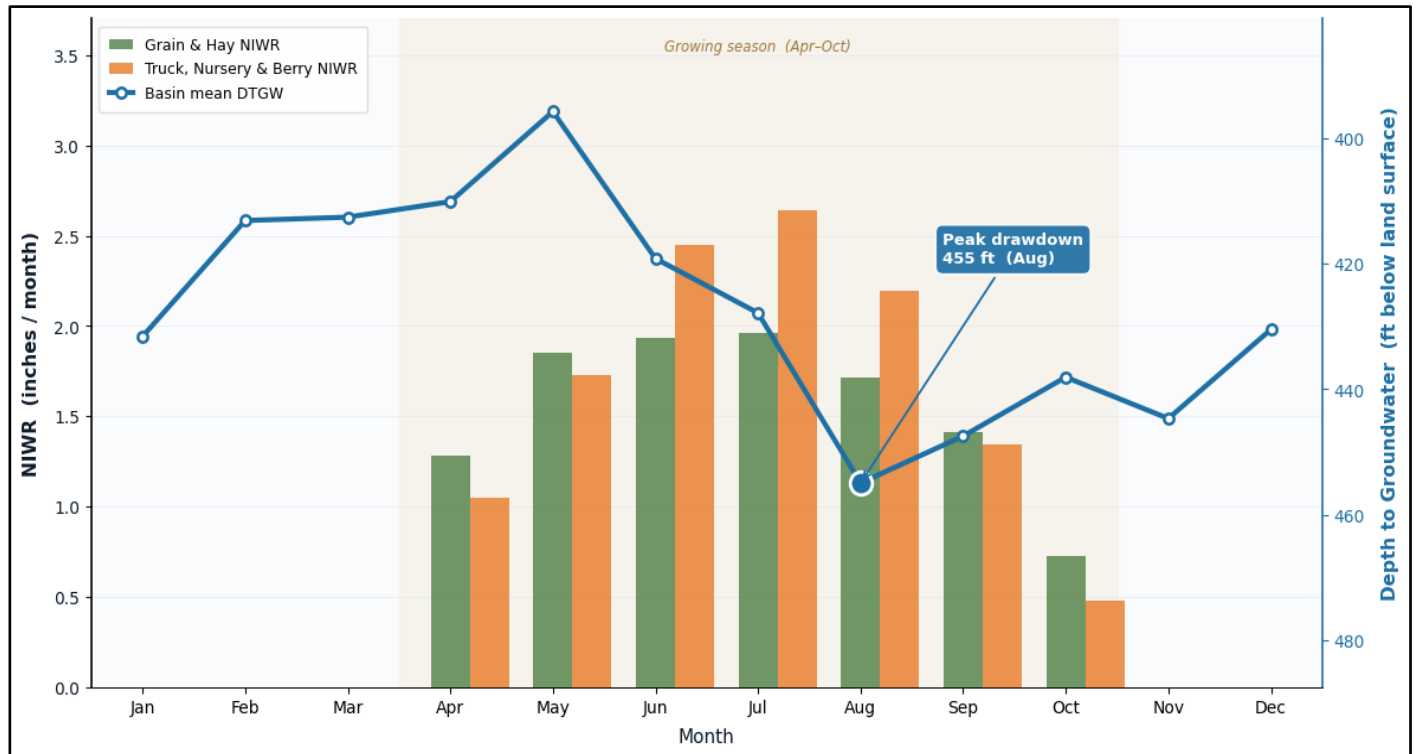


Fig 4. Dual-axis figure showing mean monthly net irrigation water requirement (NIWR, in/month) for Grain & Hay and Truck, Nursery & Berry, overlaid with mean monthly depth to groundwater (ft below land surface), averaged across 2016–2024. NIWR was computed as OpenET ET minus GRIDMET precipitation. The DTGW axis does not extend to 0 to highlight monthly fluctuations in water table in relation to irrigation demand. The relation between NIWR and DTGW reflects the seasonal water table fluctuation, providing evidence that summer irrigation demand drives aquifer drawdown (10).

Discussion

The study area map reveals that agriculture is concentrated in the central basin, where the monitoring well density is at its greatest (Fig. 1). The ET climatology reveals differences in peak water demand between crops. Grain & Hay crops peak in May–June at approximately 2.0 in/month while Truck, Nursery & Berry crops peak two months later in July, sustaining high demand through August (Fig. 2). This lag in crop growth is important because pumping later in the growing season occurs when the aquifer is already drawn down from earlier demand. Fig. 3 shows that the most overdrafted wells cluster in the central agricultural zone, directly beneath Truck, Nursery & Berry crop parcels, with depletion rates that have remained primarily linear and negative. The seasonal NIWR and DTGW figure

relates irrigation demand to aquifer response (Fig. 4). The mean DTGW throughout Central Cuyama reaches its deepest point in August at 455 ft, aligning with peak NIWR. This aligns with previous findings where seasonal groundwater level changes reflect ET-driven extraction (10, 13). Grain & Hay and Truck, Nursery & Berry show demand peaks separated by roughly two months, suggesting that the two crop types place stress on the aquifer at different points in the growing season rather than simultaneously. This concern applies to how sustainability thresholds are set. Most GSAs define minimum thresholds without meaningful stakeholder input. Economically vulnerable groups are less represented in the planning process and less protected by the final plans (14). In Cuyama, where the most water-intensive crops are grown by large agricultural corporations, a one-size-fits-all pumping framework risks disproportionately supporting dominant pumpers while overdraft continues throughout the basin.

Policy Implications

The 2025 GSP requires 60% basin-wide pumping reductions by 2040, with minimum threshold triggers when 30% of representative monitoring wells exceed their DTGW elevations for two consecutive years (2). The depletion rates reach up to 9.3 ft/yr in the central zone, suggesting several wells may exceed individual threshold before 2040. The ERA economic analysis projected that lower-value annual crops, such as wheat and carrots, would be reduced first as pumping restrictions increased, with perennial crops remaining relatively stable due to capital investment (8). The 2024 DWR crop map shows the converse. Perennial vineyard acreage has expanded substantially since 2014, including the establishment of Harvard's Brodiaea vineyard and other new plantings in the western basin (3). Perennials have a more stable demand floor than annual crops because they cannot be fallowed without destroying the initial investment, which the ERA report highlighted as a key limitation of their model (8). The NIWR analysis builds on this concern. Truck, Nursery & Berry crops generate higher and later-season irrigation demand than Grain & Hay. A pumping allocation framework that treats all pumpers equally will underestimate the contribution of high-value crops with major water demand to overdraft. An equitable and effective GSP may require crop-differentiated pumping allocations. Unless pumping allocations account for crop water intensity, the shift toward more water-intensive crops may continue to work against the basin's groundwater sustainability goals.

References

1. S. Jasechko et al., Rapid groundwater decline and some cases of recovery in aquifers globally. *Nature* **625**, 715–721 (2024).
2. Cuyama Basin Groundwater Sustainability Agency, *Cuyama Valley Groundwater Basin Final 2025 Groundwater Sustainability Plan* (Cuyama Basin Groundwater Sustainability Agency, Santa Barbara County, CA, 2025).
3. R.T. Hanson et al., Hydrologic models and analysis of water availability in Cuyama Valley, California. *U.S. Geol. Surv. Sci. Invest. Rep.* 2014–5150 (2014).
4. ERA Economics LLC, *Direct Economic Impact Analysis of the Cuyama Groundwater Basin Groundwater Sustainability Plan Demand Management Program* (Cuyama Basin Groundwater Sustainability Agency, Santa Barbara County, CA, 2019).
5. C. Walsh, Troubling trends in the battle for California's groundwater. UC Santa Barbara Division of Social Sciences (2025). <https://www.socialsciences.ucsb.edu/news/troubling-trends-battle-californias-groundwater>
6. California Department of Water Resources, Statewide Crop Mapping. California Natural Resources Agency Open Data Portal (2024). <https://data.cnra.ca.gov/dataset/statewide-crop-mapping>. Accessed March 2026.
7. N.K. Mall, J.D. Herman, Water shortage risks from perennial crop expansion in California's Central Valley. *Environ. Res. Lett.* **14**, 104014 (2019).
8. Cuyama Basin Groundwater Sustainability Agency, Data from "Cuyama Valley Basin Data Management System." <https://opti.woodardcurran.com/cuyama/main.php#>. Accessed 20 Jan 2026.
9. M. Anderson et al., OpenET: Filling a critical data gap in water management for the western United States. *J. Am. Water Resour. Assoc.* **58**, 971–994 (2022).
10. J.M. Volk et al., Assessing the accuracy of OpenET satellite-based evapotranspiration data to support water resource and land management applications. *Nat. Water* **2**, 193–205 (2024).
11. J.T. Abatzoglou, Development of gridded surface meteorological data for ecological applications and modelling. *Int. J. Climatol.* **33**, 121–131 (2013).
12. X. Hou et al., A review of advances in groundwater evapotranspiration research. *Water* **15**, 969 (2023).

13. S. Zipper et al., Estimating irrigation water use from remotely sensed evapotranspiration data: Accuracy and uncertainties at field, water right, and regional scales. *Agric. Water Manag.* **302**, 109000 (2024).
14. D. Perrone et al., Stakeholder integration predicts better outcomes from groundwater sustainability policy. *Nat. Commun.* **14**, 3793 (2023).